No connected graph with at most ten vertices is an unweighted death-Birth amplifier of weak selection: an exhaustive census, and a Lean check that Richter's eleven-vertex graph is one

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Abstract

For death-Birth (dB) updating on a connected, simple, unweighted graph G with N vertices, Allen et al. showed that $\rho_G(1+\delta) = \frac{1}{N} + \delta\,\frac{N_{\text{eff}}-2}{2N} + O(\delta^2)$, with N_{eff} an effective population size computed from coalescence times of random walks, and called G an *amplifier of weak selection* if $N_{\text{eff}} > N$. Richter (2020) exhibited an eleven-vertex example. We enumerated *every* connected graph with $3 \le N \le 10$ (2, 6, 21, 112, 853, 11 117, 261 080 and 11 716 571 graphs) and found **none** with $N_{\text{eff}} > N$: every regular graph has $N_{\text{eff}} = N$ exactly and every other graph has $N_{\text{eff}} < N$, the closest non-regular graph at $N = 10$ lying at least 0.0207 below N. In Lean 4 we check that on Richter's graph *every* rational solution of Allen's coalescence system has $N_{\text{eff}} = 2668552698098/242579025123 \approx 11.000756 > 11$. Together, *within the weak-selection criterion for unweighted graphs*, the smallest dB amplifier has exactly 11 vertices.

Tier split. [A] (Lean 4.34.1, axioms within {`propext`, `Classical.choice`, `Quot.sound`}): the theorem `R11_amplifier` (above), the uniqueness theorem `eq5_unique` (a positive solution of the system is the only solution), the isothermal control `P11_isothermal` ($N_{\text{eff}} = 11$ on the quartic parent), and kernel-decided negative controls. [L] (literature, not re-proved): the criterion $N_{\text{eff}} > N$, Allen's coalescence equations, and the step from $N_{\text{eff}} > N$ to $\rho_G > \rho_{K_N}$ for r slightly above 1 (their eq. (8)). [C]: the census itself (enumeration with nauty `geng`, counts equal to OEIS A001349; floating-point proposals decided by an exact integer residual bound; exact `Fraction` solves for the regular graphs), and exact-rational sign certificates for $\rho_G - \rho_{K_{11}}$ at four values of r. The census is *not* kernel-checked. [X]: finite-r numerics beyond the certified points (extrapolated windows, floating-point ratios). Every claim in the text carries one of these tags and a ledger identifier (`AMC-·-nnnn`).

What this is NOT. This is mathematics of graphs and of fixation probabilities in a Moran-type model, with the single-mutant, weak-selection criterion of Allen et al. It is **not** a statement about real tumours, tissues, patients or any medical application: no biological structure was studied, no data were used, and nothing here bears on diagnosis, treatment, dose or schedule. It is also not a statement about weighted or directed graphs, other update rules, or selection of finite strength.

Markers and ledger. [A]: kernel-checked Lean, axioms within the allowlist, controls caught. [L]: a cited source, taken as stated. [C]: executed code with an argument that is written down but not machine-checked. [X]: floating point, supporting nothing by itself. A [A] tag attaches to the Lean statement, never to its biological or physical gloss. Identifiers `AMC-·-nnnn` refer to `ledger/ledger.json` (Section 7).

1 Setting, predecessor and preregistration

This is item C1 of a computational programme (Proposal 4, preregistered 2026-10-27 in `results/proposal4/2026-10-27/preregistration.json`). It continues an earlier kernel-checked paper on the *isothermal theorem* for birth-death updating [9]: there, a population on an isothermal graph (equal column and row sums of the replacement matrix) fixes mutants exactly as the well-mixed population does, for every fitness r. For death-Birth updating the analogous statement is only first order: regular graphs tie with the complete graph at order δ, and the question becomes which graphs do *better* (amplifiers) or *worse* (suppressors) than a well-mixed population when selection is weak. Allen et al. [1] gave an exact first-order criterion; Richter [2, 3] and Svoboda et al. [7] describe the eleven-vertex example and call it "the smallest known unweighted dB-amplifier", without an exhaustive statement. The preregistered item C1b asked: *either no connected unweighted graph with $N \le 10$ has $N_{\mathrm{eff}} > N$, or the exact list of those that do.* The kill rules were: if the $N = 10$ census exceeds two budgets, report $N \le 9$; if the criterion's hypotheses cannot be pinned from the primary source, the census stays Tier X. Neither fired: $N = 10$ took about 24 minutes of one core, and the criterion was read in Allen et al. (arXiv v1, pp. 5, 7–8). Prior 0.45 for a definite census; the result is a definite (negative) census.

2 A step-by-step account for non-specialists

Step 1: the death-Birth process. [L] A graph G has N vertices, each occupied by one individual: a *resident* (fitness 1) or a *mutant* (fitness $r = 1 + \delta$). In one step a vertex i is chosen *uniformly at random to die*; each neighbour j of i then competes to fill the vacancy with probability proportional to its fitness, $f_j / \sum_{k \sim i} f_k$, and the winner places a copy of itself on i. Start with one mutant at a uniformly random vertex. The process ends when the mutant type has taken over (*fixation*) or died out. The *fixation probability* $\rho_G(r)$ is the chance of fixation. For the complete graph K_N it is an explicit formula; when $r = 1$ every graph has $\rho = 1/N$. (Lean: the process is *not* formalised; only the linear system of Step 3 is.)

Step 2: weak selection, and what "amplifier" means here. [L] Write $r = 1 + \delta$. Allen et al. show

$$\rho_G(1+\delta) = \frac{1}{N} + \delta\,\frac{N_{\mathrm{eff}} - 2}{2N} + O(\delta^2) \qquad \text{(their eq. (8))}, \tag{1}$$

so for K_N (where $N_{\mathrm{eff}} = N$) the slope is $(N-2)/2N$, which agrees with the exact formula $\frac{N-1}{N}\frac{1-1/r}{1-r^{1-N}}$ (a check we made). Hence $\rho_G(1+\delta) - \rho_{K_N}(1+\delta) = \delta\,(N_{\mathrm{eff}} - N)/(2N) + O(\delta^2)$. If $N_{\mathrm{eff}} > N$ then G beats K_N for all small $\delta > 0$ and loses for small $\delta < 0$: G is an *amplifier of weak selection*. If $N_{\mathrm{eff}} < N$ it is a *suppressor*. If $N_{\mathrm{eff}} = N$ the first order decides nothing. Notice what this is not: the statement concerns r close to 1 and an unknown threshold $b > 1$ beyond which the sign may flip; it does not say G amplifies for every $r > 1$.

Step 3: coalescence times and N_{eff}. [L] Follow two lineages backward in time. Each step, one of the two (chosen with probability $\frac{1}{2}$) jumps to a uniformly random neighbour: from i it goes to k with probability $p_{ik} = 1/k_i$ for $k \sim i$, where k_i is the degree. Let τ_{ij} be the expected time until they meet, with $\tau_{ii} = 0$. Then (Allen eq. (5), `AmplifierCensus.Eq5`)

$$\tau_{ij} = 1 + \tfrac{1}{2}\sum_k \big(p_{ik}\tau_{jk} + p_{jk}\tau_{ik}\big) \quad (i \ne j). \tag{2}$$

The *remeeting time* is $\tau_i = 1 + \sum_j p_{ij}\tau_{ij}$ (eq. (6), `remeet`), and with $\pi_i = k_i / \sum_j k_j$

$$N_{\mathrm{eff}} = \sum_i \pi_i \tau_i \qquad \text{(eq. (9), } \texttt{neff}\text{)}. \tag{3}$$

Worked example (exact, `data/worked_example.py`). Path on three vertices $0 - \{1, 2\}$ (a star S_2, centre 0): by symmetry $\tau_{01} = \tau_{02}$ and (2) gives $\tau_{01} = 1 + \frac{1}{4}\tau_{12}$, $\tau_{12} = 1 + \tau_{01}$, hence $\tau_{01} = \frac{5}{3}$, $\tau_{12} = \frac{8}{3}$, all $\tau_i = \frac{8}{3}$ and $N_{\mathrm{eff}} = \frac{8}{3} < 3$: a suppressor (it equals $4n/(n+1)$ at $n = 2$, Allen's star). The triangle K_3 gives all $\tau_{ij} = 2$, $\tau_i = 3$, $N_{\mathrm{eff}} = 3 = N$. The graph K_4 minus an edge gives $N_{\mathrm{eff}} = 173/45 \approx 3.844 < 4$. Two identities are used

by the code (checked exactly on all 994 connected graphs with $3 \le N \le 7$, `AMC-C-0006`): $\sum_i \pi_i^2 \tau_i = 1$ (eq. (7)), and, because $\pi_i p_{ij} = 1/2m$ for m edges,

$$N_{\text{eff}} = 1 + \frac{1}{m} \sum_{\{i,j\} \in E} \tau_{ij}. \tag{4}$$

Step 4: why regular graphs tie. [L] [C] If every degree equals k, then $\pi_i = 1/N$, and eq. (7) reads $\frac{1}{N^2} \sum_i \tau_i = 1$, so $N_{\text{eff}} = \frac{1}{N} \sum_i \tau_i = N$ exactly. This is the first-order isothermal theorem. The census confirms it for all $1 + 2 + 2 + 5 + 4 + 17 + 22 + 167$ connected regular graphs with $3 \le N \le 10$ and shows no non-regular graph attains equality (`AMC-C-0004`). In Lean, the same fact on the quartic parent of Richter's graph is `P11_isothermal` ([A], `AMC-A-0003`): $N_{\text{eff}} = 11$ exactly.

Step 5: deciding $N_{\text{eff}} > N$ for 11.7 million graphs without trusting a float. [C] Solving (2) exactly for every graph with $N = 10$ would cost hours per 10^5 graphs. Instead, a floating-point solve only *proposes*; an exact integer computation *decides*.

1. Eliminate $\tau_{ii} = 0$ and multiply (2) by $2k_i k_j$. The unknown pair-vector τ then satisfies an integer system $A_s \tau = b$, equivalently $(I - M)\tau = \mathbf{1}$ with $M \ge 0$ entrywise (entries $\frac{1}{2}p$) and row sums at most 1 (mass that lands on a diagonal pair is lost: the lineages have met).
2. Because the lineages of a connected graph do meet, $M^n \to 0$ and $(I - M)^{-1} = \sum_n M^n \ge 0$ entrywise; and $(I - M)^{-1}\mathbf{1} = \tau$ is the expected meeting-time vector.
3. Let $\tilde{\tau}$ be any approximate solution (here the float solution rounded to a multiple of 2^{-30}). Its *residual* $\rho = (I - M)\tilde{\tau} - \mathbf{1}$ is computed exactly in 64-bit integers, giving $\varepsilon = \|\rho\|_\infty \le E/2^{31}$. Then $\tau - \tilde{\tau} = -(I - M)^{-1}\rho$, and since $(I - M)^{-1} \ge 0$,

   $$|\tau_{ij} - \tilde{\tau}_{ij}| \le (I - M)^{-1}|\rho| \le \varepsilon\,(I - M)^{-1}\mathbf{1} = \varepsilon\,\tau_{\max}\ , \qquad \tau_{\max} \le \frac{\tilde{\tau}_{\max}}{1 - \varepsilon}.$$

   Non-negativity of the inverse is the whole point: it turns a residual into a *componentwise* error bound without computing any norm of the inverse.
4. By (4), $|N_{\text{eff}} - \hat{N}_{\text{eff}}| \le \varepsilon \tau_{\max}$. A graph is *excluded* (certified $N_{\text{eff}} < N$) only if $\hat{N}_{\text{eff}} + \varepsilon\tau_{\max} < N$, tested in exact integers.
5. Every graph that is not excluded is solved exactly in rational arithmetic. In the census these were precisely the regular graphs (22 at $N = 9$, 167 at $N = 10$).

The largest ε seen was 5.9×10^{-8} (at $N = 10$), while the smallest certified margin $N - N_{\text{eff}}$ among non-regular graphs was 0.0174 ($N = 9$) and 0.0207 ($N = 10$): the error budget is about six orders of magnitude below the decision margin. This argument is a hand proof implemented in Python; it is [C], not kernel-checked.

Step 6: the Lean statement beside the data. [A] Lean cannot reasonably enumerate 10^7 graphs, but it can check one graph exactly. Let `Eq5 A tau` be (2) over $\mathbb{Q}$ for the adjacency A on `Fin n`. `eq5_unique` says: *if v solves `Eq5` and $v_{ij} > 0$ for $i \ne j$, then every rational solution τ equals v* (a maximum principle: the largest ratio δ_{ij}/v_{ij} of the difference $\delta = \tau - v$ would otherwise satisfy a strict inequality). So a single rational vector, produced by untrusted code and re-checked by the kernel with `decide +kernel`, certifies N_{eff} for *all* solutions of the system, not for a guessed one. `R11_amplifier` then states

$$\forall \tau,\ \text{Eq5}\ G_{R11}\ \tau \Rightarrow N_{\text{eff}}(G_{R11}, \tau) = \tfrac{2668552698098}{242579025123}\ \wedge\ 11 < N_{\text{eff}}(G_{R11}, \tau).$$

The kernel also checks that the adjacency is symmetric, loopless, and has degrees 3 at vertices $0, 9$ and 4 elsewhere. What Lean does *not* check: that this linear system is the right one for the death-Birth process, that eq. (1) holds, or that the census is complete.

Step 7: from $N_{\text{eff}} > N$ to finite r. [L] [C] By (1) the step from $N_{\text{eff}} > N$ to "$\rho_G > \rho_{K_N}$ for $r \in (1, b)$" is literature (AMC-L-0002). To check that this is not an artefact of our reading of the criterion, we solved the exact $2^N - 2$-state chain for every connected graph with $N \leq 6$ and every 40th graph at $N = 7$ and found slope $(N_{\text{eff}} - 2)/2N$ in every case (AMC-C-0006). For the single graph with $N = 11$ we also certified signs of $\rho_G - \rho_{K_{11}}$ at chosen r with the same device as Step 5: the transient-state matrix Q of the 2046-state chain has $(I - Q)^{-1} \geq 0$ with row sums equal to expected absorption times T, so an exact residual bound gives $|x - \tilde{x}| \leq \|\rho_x\|_\infty T_{\max}$ with $T_{\max} \leq 242.43$ (AMC-C-0011).

3 Results

3.1 The census

Table 1: Connected simple graphs with $3 \leq N \leq 10$ (nauty `geng -c`), classified by N_{eff} against N (AMC-C-0001, -0002, -0004). Counts equal OEIS A001349; "$N_{\text{eff}} = N$" counts equal OEIS A005177 and all these graphs are regular. The last column is a certified lower bound on $N - N_{\text{eff}}$ over non-regular graphs ([C]).

N	graphs	$N_{\text{eff}} > N$	$N_{\text{eff}} = N$ (regular)	$N_{\text{eff}} < N$	max ε	smallest gap $N - N_{\text{eff}}$
3	2	0	1	1	4.7×10^{-10}	0.3333
4	6	0	2	4	2.8×10^{-9}	0.1556
5	21	0	2	19	1.1×10^{-8}	0.0753
6	112	0	5	107	1.0×10^{-8}	0.0612
7	853	0	4	849	1.9×10^{-8}	0.0318
8	11 117	0	17	11 100	3.3×10^{-8}	0.0330
9	261 080	0	22	261 058	4.5×10^{-8}	0.0174
10	11 716 571	0	167	11 716 404	5.9×10^{-8}	0.0207

$N = 1, 2$ are trivial (K_2 is regular, so $N_{\text{eff}} = 2$). The independent pure-`Fraction` implementation (no identity (4), no enclosure) was run on all 12 111 connected graphs with $N \leq 8$ and agrees with the enclosure census on every count and every classification (AMC-C-0003); an all-`Fraction` run at $N = 9$ was started and stopped (projected about 3 h of one core), so $N = 9, 10$ rest on the enclosure. The gap column is called `gap_upper` in the data files; it is a lower bound on $N - N_{\text{eff}}$ (a misnomer in the code).

3.2 What the nearest misses look like

Table 2: Non-regular graph closest to N (smallest certified gap), [C] observation for $N \leq 10$ (AMC-C-0005). Exact values are rational; at $N = 9, 10$ the value is a floating-point reading.

N	graph6	description	N_{eff}	gap bound
3	`BW`	path P_3	8/3	0.3333
4	`C^`	K_4 minus an edge	173/45	0.1556
5	`D]{`	K_5 minus a maximum matching	719/146	0.0753
6	`E^~w`	K_6 minus an edge	291/49	0.0612
7	`F]~vw`	K_7 minus a maximum matching	5038/723	0.0318
8	`G^~~~{`	K_8 minus an edge	2653/333	0.0330
9	`H]~v~z~`	K_9 minus a maximum matching	≈ 8.98262	0.0174
10	`I^~~~~~~w`	K_{10} minus an edge	≈ 9.97927	0.0207

The nearest non-regular graph is always *nearly complete*: for even N the complete graph minus one edge, for odd N the complete graph minus a maximum matching (one vertex keeps degree $N - 1$). It is not a sparse or a bridged graph. The sparse structures (cliques joined through bottlenecks) that Richter [3] identifies as the ingredients of amplifiers do not yet clear the bar below $N = 11$. This is an observation for $N \leq 10$; there is no proof for general N, and the descriptions are read from the graph6 strings of the data files, not proved by a separate argument.

3.3 Richter's eleven-vertex graph

Graph6 `J?ouPjI[@i?`: the unique quartic (4-regular) graph on 11 vertices (among 265) from which an *amplifier constructor* edge removal works, minus one edge at the argmax-remeeting vertex. It has 21 edges and degrees $3, 4, \ldots, 4, 3$ (the two degree-3 vertices are non-adjacent). All four edge removals at that vertex give isomorphic graphs.

Table 3: Richter's $N = 11$ graph and the cubic $N = 12$ example ([C] except the Lean row; `AMC-A-0002`, `AMC-C-0009`, `-0010`).

quantity	value here	Richter prints
N_{eff}, $N = 11$ (Lean: for *every* solution of Eq5) [A]	$2668552698098/242579025123 \approx 11.000756$	11.0008
N_{eff} of the quartic parent [A]	11 exactly	—
N_{eff}, $N = 12$ cubic example [C]	≈ 12.026644	12.0266
$\tau_{\max}$, $N = 11$ parent [C]	$10066/569 \approx 17.69069$	17.6997
$\tau_{\max}$, $N = 12$ parent [C]	≈ 18.29410	18.2911
margin $N_{\text{eff}} - N$ at $N = 11$	7.56×10^{-4}	—

The $\tau_{\max}$ values differ from the printed ones in the third decimal. The N_{eff} values agree to every printed digit. This is an **unexplained discrepancy**: it may be a typo or a convention difference; it has not been raised with the author and has no second independent check (`AMC-C-0010`).

3.4 Finite r on the eleven-vertex graph

[C] Using the Step 7 device on the 2046-state dB chain, we certified (exact-rational bounds, total error $\leq 3.3 \times 10^{-14}$): So G beats K_{11} just above $r = 1$ and loses just below it, exactly the pattern predicted by

Table 4: Certified intervals for $\rho_G(r) - \rho_{K_{11}}(r)$ on Richter's graph (`AMC-C-0011`).

r	interval	sign
$10001/10000 = 1.0001$	$[3.02059 \times 10^{-9},\ 3.02065 \times 10^{-9}]$	$+$
$2501/2500 = 1.0004$	$[7.07914 \times 10^{-9},\ 7.07919 \times 10^{-9}]$	$+$
$2499/2500 = 0.9996$	$[-2.03974 \times 10^{-8},\ -2.03973 \times 10^{-8}]$	$-$
$1001/1000 = 1.001$	$[-7.3995 \times 10^{-9},\ -7.3994 \times 10^{-9}]$	$-$

$N_{\text{eff}} > N$ ([L]), and loses again at $r = 1.001$. Bisection between $r = 1.0007$ $(+)$ and $r = 1.00085$ $(-)$ with certified signs at every step (`AMC-C-0012`) gives a certified sign change in $r \in [1.00082334, 1.00082349]$ (exact [10248431/10240000, 4099373/4096000]): We certify signs at these points only: not that there is exactly one

Table 5: The bisection (all signs certified, [C]).

r	decimal	sign	interval for $\rho_G - \rho_K$
10007/10000	1.0007	$+$	$[3.6138 \times 10^{-9}, 3.6139 \times 10^{-9}]$
20017/20000	1.00085	$-$	$[-9.4779 \times 10^{-10}, -9.4774 \times 10^{-10}]$
40031/40000	1.000775	$+$	$[1.5691 \times 10^{-9}, 1.5692 \times 10^{-9}]$
16013/16000	1.0008125	$+$	$[3.6969 \times 10^{-10}, 3.6974 \times 10^{-10}]$
160133/160000	1.00083125	$-$	$[-2.7430 \times 10^{-10}, -2.7423 \times 10^{-10}]$
320263/320000	1.00082188	$+$	$[5.1386 \times 10^{-11}, 5.1447 \times 10^{-11}]$
640529/640000	1.00082656	$-$	$[-1.1053 \times 10^{-10}, -1.1047 \times 10^{-10}]$
256211/256000	1.00082422	$-$	$[-2.9333 \times 10^{-11}, -2.9290 \times 10^{-11}]$
2562107/2560000	1.00082305	$+$	$[1.1083 \times 10^{-11}, 1.1138 \times 10^{-11}]$
5124217/5120000	1.00082363	$-$	$[-9.1128 \times 10^{-12}, -9.0596 \times 10^{-12}]$
10248431/10240000	1.00082334	$+$	$[9.829 \times 10^{-13}, 1.0493 \times 10^{-12}]$
4099373/4096000	1.00082349	$-$	$[-4.0563 \times 10^{-12}, -4.0124 \times 10^{-12}]$

crossing, and not the sign between them. Richter reads $r_{\max} \approx 1.00075$ off a figure; our bracket is a different,

certified quantity ([X] for any comparison with the figure). Beyond these points the "window $(1, 1.0008)$" is an [X] extrapolation (`AMC-X-0001`).

3.5 Regular graphs at finite r

[C] For all 212 connected non-complete regular graphs with $4 \le N \le 10$ we certified $\rho_G(r) < \rho_{K_N}(r)$ at both $r = 999/1000$ and $r = 1001/1000$ (none above, none undecided; `AMC-C-0013`). This is consistent with these ties being second-order *reducers* of fixation near $r = 1$, but it is **not** a proof for all r close to 1; Allen et al. (p. 21) list it as open.

4 The headline, and its tiers

statement	tier	basis
$N_{\text{eff}} > N$ iff amplifier of weak selection; first-order expansion (1)	[L]	Allen et al., arXiv v1, pp. 5–9, 32–34 (`AMC-L-0001`, `-0002`)
Any rational solution of Eq5 on G_{R11} has $N_{\text{eff}} = 2668552698098/242579025123 > 11$	[A]	`R11_amplifier` (`AMC-A-0002`)
A positive solution of Eq5 is the only solution	[A]	`eq5_unique` (`AMC-A-0001`)
Quartic parent has $N_{\text{eff}} = 11$	[A]	`P11_isothermal`
No connected graph with $3 \le N \le 10$ has $N_{\text{eff}} > N$; regular $\Leftrightarrow N_{\text{eff}} = N$	[C]	geng + OEIS counts + integer enclosure (`AMC-C-0001..0004`)
The smallest connected unweighted dB amplifier of weak selection has exactly 11 vertices	[C]	census (C) + Lean existence (A) + criterion (L) (`AMC-C-0014`)
Sign pattern of $\rho_G - \rho_{K_{11}}$ at four r, and a sign change in $[1.00082334, 1.00082349]$	[C]	exact-rational residual bounds (`AMC-C-0011`, `-0012`)
Extrapolation of windows; floating ratios	[X]	`AMC-X-0001`

The headline is filed Tier C because its census half is. The ledger gate would flag any claim filed above the tier of something it rests on.

5 Controls

Every control was run before any census count was reported (`data/controls.json`, `data/logs/controls.log`; `AMC-C-0006..0009`, `AMC-A-0003/0004`).

Positive controls (all met).

- K_N and C_N, $N = 3..10$: $N_{\text{eff}} = N$ exactly (Allen's weak-selection isothermal theorem).
- All regular graphs have $N_{\text{eff}} = N$ exactly; their counts $1, 2, 2, 5, 4, 17, 22, 167$ equal OEIS A005177, so none was missing.
- Star S_n, $n = 2..9$: $N_{\text{eff}} = 4n/(n+1)$ exactly.
- Slope control: $d\rho/dr|_{r=1}$ of the exact 2^N-state chain equals $(N_{\text{eff}} - 2)/(2N)$ on every connected graph with $N = 3..6$ and on 163 graphs (every 40th) at $N = 7$.
- Closed forms against the exact chain at $r \in \{\frac{3}{4}, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, \frac{1001}{1000}\}$, $N = 3..7$: dB and Bd on K_N; the dB cycle as given by Kaveh et al. eq. (20) at $d = 1$, Hindersin–Traulsen SI eq. (18) and Richter eq. (12), namely $2(r-1)/\big(3r - 1 + (r-3)r^{2-N}\big)$; the dB star (Allen eq. (2)).
- Hindersin & Traulsen, $N = 4$: their C_4 dB eqs. (13), (14) reproduced exactly; the four degree-heterogeneous graphs are Bd amplifiers and dB suppressors; C_4 is the only dB-"unclassified" graph; K_4 equals the well-mixed population. Stars $S_4..S_6$ are Bd amplifiers and dB suppressors on the r grid.
- Richter: among the 265 quartic graphs on 11 vertices exactly one parent is an amplifier constructor, all four removals at its argmax vertex give $N_{\text{eff}} = 11.000756$ (printed 11.0008); among the 85 cubic graphs on 12 vertices exactly one constructor, two removals give 12.026644 (printed 12.0266; the third removal disconnects).

Negative controls (all rejected as intended).

- **Allen eq. (3) as printed in arXiv v1.** The cycle formula with $r^{-(N-1)}$ *disagrees with the exact chain in 25 of 25 cases*; three other sources and the exact chain agree with $r^{2-N}(r-3)$. We report the disagreement and do not assert whose error it is; the criterion used here (eqs. (5)–(9)) does not depend on eq. (3), and the point has not been raised with the authors (`AMC-C-0007`).
- A planted wrong random walk, $p_{ij} = 1/k_j$ in place of $1/k_i$, breaks the star value $10/3$.
- K_N never passes the strict test $N_{\text{eff}} > N$.
- The Antal approximation $N\mu_1^2/\mu_2$ is $\leq N$ on every connected graph with $N \leq 7$, so it can never detect an amplifier; it "fails" the Richter control by construction.
- Edges removed away from the argmax vertex: $N_{\text{eff}} \leq 10.8805 < 11$ on Richter's parent, $\leq 11.9337 < 12$ on the cubic parent.
- Lean (kernel decides the right way, `AMC-A-0004`): `ctl_perturbed_vector_rejected` (one τ entry changed by $1/D$ gives ¬Eq5), `ctl_perturbed_graph_rejected` (the R11 certificate is ¬Eq5 on the parent: a wrong transition matrix is caught), `ctl_parent_not_amplifier`.
- Lean (caught by the axiom audit): the planted `ctl_sorry` reports `sorryAx` and `ctl_axiom` reports `AmplifierCensusControls.ctl_cheat`, although `Controls.lean` compiles with rc 0 (a compile exit code alone accepts both).
- `ControlsMustFail.lean` asks the kernel to accept the perturbed vector and fails with rc 1; the error text shows elaborator-side reduction diagnostics rather than "evaluates to false", so the decisive evidence is the kernel-checked negation above.
- A scan for `skipKernelTC`, `debug.`, `addDecl`, `#exit`, `implemented_by`, `native_decide` and `set_option` in the module is clean.

6 Lean formalisation

Modules in `formal_cert/Programmes/AmplifierCensus/` (copies in `data/lean/`): `Core.lean` (definitions `deg`, `p`, `Eq5`, `remeet`, `neff`; `eq5_unique`; `neff_of_cert`), `DataR11.lean` and `DataP11.lean` (integer certificate data proposed by the untrusted `gen_lean.py`), `Richter11.lean` (`R11_amplifier`, `P11_isothermal`, symmetry/degree checks), `Controls.lean`, `ControlsMustFail.lean` (nothing imports the last two). Imports: `Mathlib.Tactic` and `Mathlib.Algebra.BigOperators.Fin`; never `import Mathlib`; no `sorry` outside the planted control. Compiled through `scripts/certified_numerics/lean_pinned.py` (Lean 4.34.1, Mathlib `d13f23b7`). The in-file `#print axioms` reports, for `R11_amplifier`, `P11_isothermal` and `eq5_unique`, exactly [`propext`, `Classical.choice`, `Quot.sound`] (`data/logs/lean_richter11.json`); a `leanchecker` replay of `Richter11` and `Controls` returned rc 0 (`data/logs/leanchecker_replay.json`; replay does not check axioms, the audit does). This used the pinned wrapper and not `anse/formal/lean_runner.py`, a deviation from the repository's usual route. sha256 of the files:

- `Core.lean`: `df9e30617ec88468cb18d986c2606665612bc528d19335fd10db07528f75b336`
- `DataR11.lean`: `050f91ac06bb7ba122c02c25e1b8295aa0c034b72588df740e24ac380ca66571`
- `DataP11.lean`: `cac552f026aec1cf36ebf91a010e41f6f03fc3bdf3747304b88f481bb5dff67b`
- `Richter11.lean`: `7d79537ab4e16f26c4a821f0c04a6935d97acbbf61c790bdc4a1cb7f740fa1ce`
- `Controls.lean`: `8f221b83258756f6d7ad6b7975f73da30d27d115797c0971f10dee9272fdf177`
- `ControlsMustFail.lean`: `d1230dd1bdae30c2840d165230807ef7f6d2b46d4c888310cc59acccde89bd3b`

What Tier A certifies: the stated linear-system theorems on the stated adjacency. **It does not certify** that eqs. (5), (6), (9) are Allen's equations as printed (a person must compare), that $N_{\text{eff}} > N$ implies amplification ([L]), the enumeration, or any census count.

7 Claim ledger

`ledger/build_ledger.py` files 22 claims: 4 at Tier A (`AMC-A-0001..0004`), 2 at L (`AMC-L-0001, 0002`), 14 at C (`AMC-C-0001..0014`, the last being the headline) and 2 at X (`AMC-X-0001`, the extrapolation; `AMC-X-0002`, recording that the repository guards were not run). Each has a content-addressed evidence blob in `ledger/evidence/`. In the data, some finite-r statements are exact-rational certificates and are filed C as in the run note; the extrapolation beyond the certified points is X. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`; verbatim in `ledger/gate_output.txt`) exits with code **1**: **0 blocking findings and 4 flags** (`LEDGER_UNAUDITED_TIER_A`, one for each Tier A row: no *person* has certified that the Lean statements mean what their glosses say). Exit code 1 is the honest state, not a failure to be cleared by editing. There is no AI audit either, and no statement lock.

8 Limitations

- **Novelty of the exhaustive statement is unchecked beyond the papers read.** We read Allen et al. (arXiv 1906.01036v1), Richter (2008.01446v1; 2204.12861v1, in pages), Tkadlec et al. (1906.02785v1), Hindersin & Traulsen (1504.03832v2), Kaveh et al. (1411.4556v1), and, for novelty only, Svoboda et al. (2401.14914v1) and Brewster et al. (2605.26411v3, Bd only), plus OEIS A001349 and A005177. Richter [3] (p. 8) reports only that his iterative *regular-input* procedure found none with $N \leq 10$; Svoboda et al. (p. 5) call Richter's graph "the smallest known". We found no exhaustive all-graph dB statement in these eight papers, but we did not search further, and Richter's ref. [6] (all graphs with $N \leq 10$, Bd) was not identified. Nothing here should be read as a priority claim.
- **The census is Tier C, not Lean.** Completeness depends on nauty `geng` (2.7r3, Debian, Apache-2.0) and on the match with OEIS A001349; the enclosure argument is a hand proof implemented in Python; the Lean theorems cover only the $N = 11$ linear system.
- **Scope of the statement.** Unweighted, undirected, simple, connected graphs; death-Birth updating; uniform initialisation (one mutant at a uniformly random vertex); the weak-selection criterion $N_{\text{eff}} > N$ at first order in δ. Temperature or mutation-at-birth initialisation can change the classification (Hindersin & Traulsen, Richter 2022) and nothing transfers. Weighted transient amplifiers (Allen's Fans, Separated Hubs) are out of scope. A weak-selection amplifier is not an amplifier at finite r and not a universal amplifier (Tkadlec et al., Thm. 1, as recorded in the run note, forbids that for non-complete graphs).
- **Finite-r checks are not proofs for all r near 1.** The four signs, the bisection and the 212 regular graphs are certified only at the listed r; uniqueness of the crossing is not shown; whether regular graphs are reducers for all r near 1 remains open.
- **No independent audit.** No human or AI auditor has reviewed this paper, the Lean statements or the census code; the producer $\neq$ verifier recompile is pending; no statement lock was registered for the new module.
- **Repository guards were not run** (`antigravity_guard.py`, `test_rigor_guard.py`, `pytest tests/`); no guard output is claimed.
- **Deviations from the preregistration.** (1) The $N = 9, 10$ census is a certified enclosure plus exact solves of the candidates, not an exact rational solve of every graph. (2) The preregistration expected a Tier A certificate per listed amplifier; there is none at $N \leq 10$, so Tier A went to Richter's graph and its parent. (3) Finite-r certificates use Python `Fraction`s, not python-flint (not installed), so they are Tier C, as preregistered for that route. (4) C1a (the K_N and cycle closed forms) was checked as exact rational comparisons with the exact chain, not as Lean theorems. (5) C1x (birth/death-rate planes on tissue-like graphs) was **not attempted**. (6) The Lean gate used `lean_pinned.py`. (7) CPU accounting: at most two busy processes, but idle wrapper processes at times made three; some shell-guard lapses occurred (wait loops, one empty heredoc, one stray variable); none touched git.
- **Open discrepancies.** Allen eq. (3) (Section 5) and Richter's printed $\tau_{\max}$ values are unexplained and unconfirmed with the authors.
- **Data note.** The run note quotes the cubic $\tau_{\max}$ as 18.29406; the data file gives 18.29410, used here.

9 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: `25eacb7c37b7ebb54b53792cae13283a493e7bdf`; the C1 outputs are uncommitted working-tree files at that time (nothing was pushed or published). Copies of the inputs are in `data/` (`data/code/` the scripts, `data/lean/` the Lean sources, `data/logs/` compile and census logs, `data/RESULT.md` and `data/result.json` the run note). The graph6 files of the census (10^7 graphs at $N = 10$) are not included; their sha256 values are in `data/result.json`. Commands (from the repository root, the C1 run):

```
nauty-geng -c N | python -I code/census_enclosure.py N        # N = 3..10: enclosure census; then code/finish_candidates.py
python -I code/run_small.py ; python -I code/exact_census.py   # independent Fraction census, N <= 8
python -I code/controls.py ; python -I code/richter.py         # controls, Richter reproduction
python -I code/fixation_cert.py ; python -I code/window_bisect.py ; python -I code/regular_check.py
python -I code/gen_lean.py                                     # proposes DataR11.lean / DataP11.lean
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/AmplifierCensus/Richter11.lean
python -I data/worked_example.py data/worked_example.json      # the exact examples of Step 3
python -I ledger/build_ledger.py ; python -I ledger/run_gate.py   # gate: exit 1 = 0 blocking, 4 flags
```

The exact argument lists of the census scripts are in the run note and the scripts' headers; they were not re-run for this paper. The run used cores 4 and 5 (taskset) and no GPU; the $N = 10$ enclosure took 1237 s on one core.

References

[1] B. Allen, C. Sample, R. Jencks, J. Withers, P. Steinhagen, L. Brizuela, J. Kolodny, D. Parke, G. Lippner, Y. A. Dementieva, *Transient amplifiers of selection and reducers of fixation for death-Birth updating on graphs*, PLoS Comput. Biol. 16(1):e1007529 (2020), arXiv:1906.01036 (v1 read). Source of the criterion $N_{\mathrm{eff}} > N$ ([L]).

[2] H. Richter, arXiv:2008.01446 (2020), the eleven-vertex graph (Fig. 1a) and the cubic twelve-vertex example. Title not re-verified for this paper.

[3] H. Richter, arXiv:2204.12861 (2022). Title not re-verified for this paper.

[4] V. Tkadlec, A. Pavlogiannis, K. Chatterjee, M. A. Nowak, arXiv:1906.02785 (2019). Title not re-verified for this paper.

[5] L. Hindersin, A. Traulsen, *Most undirected random graphs are amplifiers of selection for Birth-death dynamics, but suppressors of selection for death-Birth dynamics*, PLoS Comput. Biol. 11(11):e1004437 (2015), arXiv:1504.03832.

[6] K. Kaveh, N. L. Komarova, M. Kohandel, *The duality of spatial death-birth and birth-death processes and limitations of the isothermal theorem*, R. Soc. Open Sci. 2:140465 (2015), arXiv:1411.4556.

[7] J. Svoboda, S. Joshi, J. Tkadlec, K. Chatterjee, arXiv:2401.14914 (2024), read for novelty only. Title not re-verified.

[8] Brewster, Lippner, Tkadlec, Nowak, arXiv:2605.26411 (v3), Bd only, read for novelty only. Title not re-verified.

[9] X. Callens, *The isothermal theorem of evolutionary graph theory for birth-death updating, kernel-checked in Lean 4: Moran fixation on strongly connected isothermal graphs and its stationary-selection generalization*, Zenodo, doi:10.5281/zenodo.23660071 (2026). Own preprint, the predecessor of this one; not peer reviewed.

[10] B. D. McKay, A. Piperno, *Practical graph isomorphism, II*, J. Symb. Comput. 60 (2014) 94–112; nauty/`geng` 2.7r3 used.

[11] OEIS Foundation, sequences A001349 (connected graphs on n nodes) and A005177 (connected regular graphs), `https://oeis.org`, read 2026-10-27.

[12] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019); cited for the library only.